

Frequency-Adaptive Sharpness Regularization for Improving 3D Gaussian Splatting Generalization

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Abstract

Despite 3D Gaussian Splatting (3DGS) excelling in most configurations, it lacks generalization across novel viewpoints in a few-shot scenario because it overfits to the sparse observations. We revisit 3DGS optimization from a machine learning perspective, framing novel view synthesis as a generalization problem to unseen viewpoints—an underexplored direction. We propose Frequency-Adaptive Sharpness Regularization (FASR), which reformulates the 3DGS training objective, thereby guiding 3DGS to converge toward a better generalization solution. Although Sharpness-Aware Minimization (SAM) similarly reduces the sharpness of the loss landscape to improve generalization of classification models, directly employing it to 3DGS is suboptimal due to the discrepancy between the tasks. Specifically, it hinders reconstructing high-frequency details due to excessive regularization, while reducing its strength leads to under-penalizing sharpness. To address this, we reflect the local frequency of images to set the regularization weight and the neighborhood radius when estimating the local sharpness. It prevents floater artifacts in novel viewpoints and reconstructs fine details that SAM tends to oversmooth. Across datasets with various configurations, our method consistently improves a wide range of baselines. Code will be available at <https://bbangsik13.github.io/FASR>.

1. Introduction

Reconstructing 3D scenes from multi-view 2D images has been a long-standing problem of interest. 3D Gaussian Splatting (3DGS) [23] achieves photo-realistic fidelity in novel view synthesis with real-time rendering. However, it requires densely captured input views, which are costly to obtain. In sparse-view settings, they often *overfit to training views*, resulting in *poor generalization to novel viewpoints* with unresolved details and floating artifacts. To improve

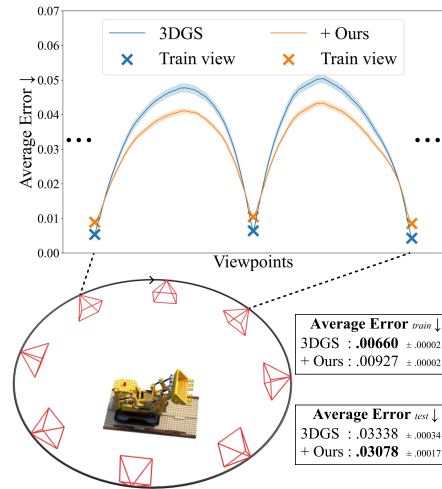


Figure 1. **Overview.** Our proposed optimization algorithm improves generalization. Given eight training views rendered from the `lego` scene in Blender synthetic dataset [41], our method maintains low Average Error [44] across interpolated novel views, whereas 3DGS exhibits overfitting. Plots are means and standard deviations over ten runs.

generalization, previous approaches have adopted various strategies, such as integrating geometric priors from depth and flow estimators [8, 66], and leveraging dense correspondence prediction models [20]. Although these methods show effectiveness, 3DGS optimization has been underexplored as a machine learning problem where generalization to novel viewpoints is important.

In this paper, we propose an optimization algorithm for 3DGS variants to improve quality in novel viewpoints, i.e., generalization (Fig. 1). Our motivation is to optimize the model toward the flat minima in the loss landscape, which are widely known to promote better generalization. As illustrated in Fig. 2, sharp minima¹ exhibit a large gap between train and test loss, whereas flat minima show a much

¹We omit “local” for brevity.

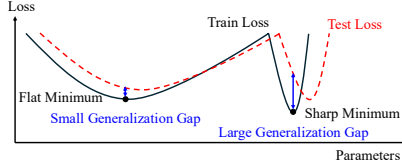


Figure 2. **Conceptual 1D Loss Landscape of Flat and Sharp Minima**³. Flat minimum better generalize than sharp minimum.

smaller gap, indicating better generalization. For example, floaters are invisible in training views but critically emerge in novel viewpoints; i.e., the Gaussians overfit to the training views. This large generalization gap implies a sharp minimum, i.e., loss dramatically changes even with slight perturbations on the model parameters².

Although Sharpness-Aware Minimization (SAM) [12] theoretically and empirically demonstrates that pursuing flatter minima leads to improved generalization of classification networks, *achieving flat minima does not always guarantee better generalization in reconstruction tasks* because the importance of sharpness for accurate reconstruction varies across regions. In particular, model parameters representing high-frequency details (e.g., edges) inherently induce a sharp loss landscape; even small changes in a well-fitted Gaussian cause a drastic change in the loss. In contrast, for parameters representing low-frequency regions, the loss changes more gradually, favoring flat minima. In this sense, finding a flat minimum for all Gaussians is problematic; sharpness in high-frequency areas is desirable for accurate reconstruction, flatter minima in low-frequency regions are preferable for better generalization.

To this end, we propose Frequency-Adaptive Sharpness Regularization (FASR), an optimization algorithm that penalizes local sharpness, where both the neighborhood radius used for its computation and the regularization weight are set in inverse proportion to the local frequency, *encouraging a flatter loss landscape while retaining the sharpness required for fine details*. We show that the enhancement of our method is complementary to prior methods, thereby providing additional performance gains across them.

Our contributions are significant as follows:

- To the best of our knowledge, we present the first fundamental investigation into the relationship between loss landscape and generalization in novel view synthesis.
- Consequently, we propose an optimization algorithm for 3DGS by reformulating SAM in a frequency-adaptive manner, overcoming the limitation of SAM, which over-smooths high-frequency details.
- Our method is versatile in improving a wide range of baseline methods on various datasets.

²Camera perturbations can be interpreted as Gaussian extrinsic parameter perturbations, e.g., parallel movement of all Gaussians is equivalent to moving the camera.

2. Related work

2.1. Sparse-view reconstruction

Reconstructing scenes from highly sparse inputs remains a fundamental challenge, as limited supervision often leads models to overfit training views, resulting in insufficient generalization in unseen views. Early efforts sought to extend neural radiance fields (NeRFs) [41] by incorporating additional regularizations or auxiliary cues. For example, these studies [9, 44, 54, 60] introduce geometry- or depth-based constraints and frequency regularization to alleviate the underconstrained nature of sparse-view optimization. Although these approaches demonstrate effectiveness, they inherit the slow training and rendering processes of NeRF.

Recent studies have shifted to use 3D Gaussian Splatting (3DGS) [23], aiming for real-time rendering efficiency. These studies [8, 20, 31, 58, 66, 68] similarly leverage external priors such as depth [6, 46], correspondence [30], or flow [49], while another focuses on ensemble-like regularization [45, 63, 65]. Sharing a key insight with our method, Sparfels [21] aims to minimize a worst-case loss for robustness; however, it does so by freezing the Gaussian means and approximating the objective with an upper bound, which simplifies to a color variance regularization along each ray. Although this proxy effectively improves details, it may overlook color-matched floaters. In contrast, our method directly optimizes the worst-case loss with respect to all Gaussian parameters, interpreting it as a sharpness regularization.

Importantly, our optimization algorithm is complementary: rather than altering the representation or adding priors, we fundamentally guide 3DGS to converge on a general solution in under-constrained sparse supervision.

2.2. Flatness and generalization

The relationship between the flatness of the loss landscape and model generalization has been extensively investigated in prior research. Keskar et al. [24] shows that converging to sharp minima leads to poor generalization, while Jiang et al. [22] identifies that sharpness is the most correlated indicator of generalization. Subsequent work [7, 18] demonstrated that averaging parameters along training trajectories can lead to flatter minima with better generalization.

Beyond averaging strategies, Sharpness-Aware Minimization (SAM) [12] explicitly estimates and reduces sharpness, inspiring subsequent work [2, 34, 42, 51] to analyze and improve upon its formulation. Moreover, some works have analyzed the accuracy of estimated sharpness as a measure of generalization. They show that sharpness

³Figure is reproduced from Keskar et al. [24]. As empirically shown in Izmailov et al. [18], the test loss landscape tends to shift relative to the train loss landscape, while Liu et al. [37] showed that the loss landscape differs with the imbalance level of the dataset.

changes with parameter rescalings [10], and the appropriate sharpness estimation differs across different training setups and tasks [3], which hinders the correlation between sharpness and generalization. Consequently, recent work introduces normalization and invariant formulations to appropriately estimate sharpness, enabling a more reliable link between sharpness and generalization [16, 19, 26, 28, 53]. On the other hand, some works report counterexamples where sharper models generalize well, indicating that flatter minima are not always the optimal strategy for generalization [4, 10, 56].

Building on this perspective, we revisit the loss landscape sharpness in the context of the reconstruction task. We hypothesize that the optimal sharpness is not uniform but varies with the local frequency of the signal. Furthermore, the radius for estimating the local sharpness should vary with the local frequency. Therefore, instead of pursuing a universally flat minimum, we introduce a frequency-adaptive optimization strategy that appropriately regularizes sharpness for the reconstruction task, allowing sharp minima where they benefit high-frequency details.

2.3. Reconstructing with perturbation

Random perturbation has been introduced into radiance field training for two primary purposes.

Unlike our approach, which aims to improve generalization, some studies employ perturbation for uncertainty quantification rather than relying on a deterministic approach. Stochastic or Bayesian formulations of NeRF have explicitly modeled radiance or density distributions [29, 48]. Subsequent work perturbs trained models to estimate epistemic uncertainty [15]. Similar ideas have been extended to 3DGS, where Gaussian parameters are sampled from learned distributions to render both images and calibrated uncertainty maps [1].

Some works [13, 25, 36] achieve robustness by injecting random noise into Gaussian parameters or query locations, which can be interpreted as randomly choosing parameters in a neighborhood radius. Nevertheless, this random perturbation serves as an inefficient proxy for finding the worst-case loss and may also introduce unexpected artifacts (Sec. C provides more details). In contrast, our method adversarially perturbs parameters along the gradient direction to calculate the worst-case loss within that radius.

3. Method

We first provide preliminary on SAM [12] (Sec. 3.1). Then, we explore applying SAM to 3DGS (Sec. 3.2). Finally, Sec. 3.3 presents our frequency-adaptive sharpness regularization.

3.1. Preliminary: Sharpness-Aware Minimization

Sharpness-Aware Minimization (SAM) [12] is an optimization method that improves generalization by encouraging solutions to lie in flat regions of the loss landscape. Along with minimizing the empirical loss at a parameter point $\mathbf{w}$, SAM estimates the loss sharpness within a neighborhood of radius ρ . The optimization then jointly minimizes both the empirical loss and this estimated sharpness:

$$L^{\text{SAM}} \triangleq \overbrace{\max_{\|\epsilon\|_2 \leq \rho} L(\mathbf{w} + \epsilon) - L(\mathbf{w})}^{\text{loss sharpness}} + \overbrace{L(\mathbf{w})}^{\text{empirical loss}} \quad (1)$$

$$= \max_{\|\epsilon\|_2 \leq \rho} L(\mathbf{w} + \epsilon),$$

worst-case loss

which is equivalent to the worst-case loss. In practice, it is approximated via first-order Taylor expansion for the loss function around the current parameters, *perturbing the parameters* in the gradient direction with magnitude ρ :

$$\hat{\epsilon}(\mathbf{w}) \triangleq \arg \max_{\|\epsilon\|_2 \leq \rho} L(\mathbf{w} + \epsilon) \approx \rho \cdot \frac{\nabla_{\mathbf{w}} L(\mathbf{w})}{\|\nabla_{\mathbf{w}} L(\mathbf{w})\|_2}. \quad (2)$$

Then, the model parameters $\mathbf{w}$ are updated via a Stochastic Gradient Descent (SGD) [43] step, with the gradient computed at the estimated local maximum $\mathbf{w} + \hat{\epsilon}(\mathbf{w})$:

$$\mathbf{w} \leftarrow \mathbf{w} - \lambda_{\text{lr}} \cdot \nabla_{\mathbf{w}} L(\mathbf{w})|_{\mathbf{w} + \hat{\epsilon}(\mathbf{w})},$$

where λ_{lr} is a learning rate. Supporting the intuition of Fig. 2, this method improves generalization by tightening the generalization bound based on sharpness derived from the PAC-Bayesian framework [11, 39, 47]:

Theorem 1. For any $\rho > 0$, with training set $\mathcal{S}$ from data distribution $\mathcal{D}$,

$$L_{\mathcal{D}}(\mathbf{w}) \leq \max_{\|\epsilon\|_2 \leq \rho} L_{\mathcal{S}}(\mathbf{w} + \epsilon) + h(\|\mathbf{w}\|_2^2 / \rho^2),$$

where $h : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is a strictly increasing function (under some technical conditions on $L_{\mathcal{D}}(\mathbf{w})$).

In practice, ρ is a hyperparameter.

The sharpness can be interpreted as a regularization term, as in Weighted SAM (WSAM) [61]:

$$L^{\text{WSAM}} \triangleq L(\mathbf{w}) + \frac{\gamma}{1 - \gamma} [\max_{\|\epsilon\|_2 \leq \rho} L(\mathbf{w} + \epsilon) - L(\mathbf{w})] \quad (3)$$

$$= \frac{1 - 2\gamma}{1 - \gamma} L(\mathbf{w}) + \frac{\gamma}{1 - \gamma} \max_{\|\epsilon\|_2 \leq \rho} L(\mathbf{w} + \epsilon).$$

When the weight hyperparameter γ is set to 0, the sharpness term vanishes and the optimization minimizes the empirical loss only. When $\gamma = 0.5$, WSAM is equivalent to SAM (Eq. (1)), while for $0.5 < \gamma < 1$, the sharpness term

Algorithm 1 Applying SAM to 3DGS

Input: Multi-view images $\tilde{I}^v$, where camera $v \in \mathcal{V}$
Output: Optimized $\mathcal{G} = (\mu, \mathbf{q}, \mathbf{s}, \sigma, \mathbf{Y}^{\text{DC}}, \mathbf{Y}^{\text{AC}})$

```

1: while  $\mathcal{G}$  not converged do
2:    $L \leftarrow \text{Loss}(\text{Render}(\mathcal{G}, v), \tilde{I}^v)$  // Get loss
3:   for all  $\theta \in \mathcal{G}$  do
4:      $\hat{\theta} \leftarrow \theta + \rho_{\theta} \frac{\nabla_{\theta} L}{\|\nabla_{\theta} L\|_2}$  // Ascent step
5:   end for
6:    $\hat{\mathcal{G}} \leftarrow (\hat{\theta} \mid \theta \in \mathcal{G})$  // Local maximum
7:    $\hat{L} \leftarrow \text{Loss}(\text{Render}(\hat{\mathcal{G}}, v), \tilde{I}^v)$  // Get loss
8:    $\mathcal{G} \leftarrow \mathcal{G} - \text{Adam}(\nabla_{\hat{\mathcal{G}}} \hat{L})$  // Descent step
9: end while
  
```

is emphasized. Accordingly, in the extended Theorem 1, $\max_{\|\epsilon\|_2 \leq \rho} L_{\mathcal{S}}(\mathbf{w} + \epsilon) = L_{\mathcal{S}}^{\text{SAM}}$ becomes $L_{\mathcal{S}}^{\text{WSAM}}$, and the function h is modified.

See Foret et al. [12] and Yue et al. [61] for the full theorem statement and proof.

3.2. Applying SAM to 3DGS

Let $\mathcal{G}_i$ be the i -th Gaussian, defined as $\mathcal{G}_i = (\mu_i, \mathbf{q}_i, \mathbf{s}_i, \sigma_i, \mathbf{Y}_i^{\text{DC}}, \mathbf{Y}_i^{\text{AC}})$, where $\mu, \mathbf{q}, \mathbf{s}, \sigma$, and $\mathbf{Y}$ represent the mean, rotation, scale, opacity and spherical harmonics (SH) coefficients, respectively, with $\mathbf{Y}^{\text{DC}}$ and $\mathbf{Y}^{\text{AC}}$ corresponding to the DC and AC terms of SH.

As in the 3DGS [23] pipeline, we render 3D Gaussians $\mathcal{G} = (\mathcal{G}_i)_{i=1}^N$ from the camera view v and compute the loss L (Algorithm 1.2). The gradient of this loss guides the parameter to its local maximum (Algorithm 1.3-6). Similar to the SAM pipeline, we then evaluate the loss $\hat{L}$ at the local maximum and use this gradient to update the original parameters via the Adam optimizer [27] (Algorithm 1.7-8).

However, this direct application often leads to suboptimal performance as shown in Sec. 4.2.1. Unlike the classification task, in the reconstruction task, the curvature of the landscape is highly correlated with the image frequency. Loss in high-frequency regions requires sharp minima for accurate reconstruction, making it sensitive to perturbations, whereas low-frequency regions require flat minima and are less sensitive to them. Due to this differing sensitivity, a fixed ρ_{θ} is problematic. Specifically, in high-frequency regions, it leads to inaccurate sharpness estimation by causing large first-order approximation errors (Eq. 2). Conversely, in low-frequency regions, the perturbation is too weak to be meaningful, limiting the improvement of SAM. Moreover, SAM penalizes estimated sharpness with the fixed regularization weight $\gamma = 0.5$. It over-penalizes the high-frequency regions, which must remain sharp to preserve details. Concurrently, it under-penalizes the low-frequency regions, failing to sufficiently improve generalization.

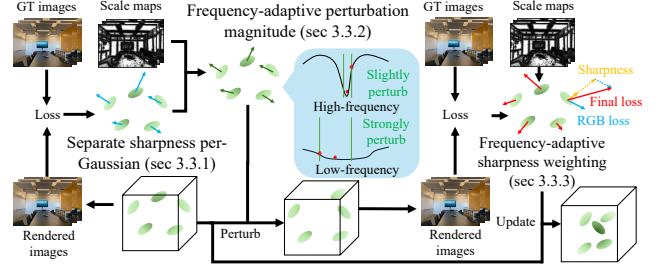


Figure 3. Overview of our proposed method.

Key hypothesis. Based on this intuition, we hypothesize that for each Gaussian, the optimal neighborhood radius ρ_{θ} and regularization weight γ vary in correlation with image frequency, which effectively tightens the WSAM-extended version of the generalization bound in Theorem 1.

3.3. Frequency-Adaptive Sharpness Regularization

To address the limitation of SAM, we introduce Frequency-Adaptive Sharpness Regularization (FASR). As illustrated in Fig. 3, we first estimate the local sharpness of each Gaussian attribute independently (Sec. 3.3.1). Then we adjust the perturbation magnitude and the regularization weight of each Gaussian attribute⁴ (Secs. 3.3.2 and 3.3.3) using the precomputed scale map Γ^v at view v , which is inversely proportional to the local frequency (detailed in Sec. E).

3.3.1. Separate sharpness per-Gaussian

Unlike neural networks, 3DGS is an explicit model, which allows us to associate the local frequency of each pixel with its corresponding Gaussian. To apply frequency adaptivity, we begin by computing gradient on the per-Gaussian attribute $\theta_i \in \mathcal{G}_i$ separately rather than the entire parameter set. Accordingly, Algorithm 1.4 becomes

$$\hat{\theta}_i \leftarrow \theta_i + \rho_{\theta} \frac{\nabla_{\theta_i} L}{\|\nabla_{\theta_i} L\|_2}. \quad (4)$$

This step has the advantage of estimating the local sharpness of each Gaussian attribute independently. Specifically, it mitigates the first-order approximation error of sharpness by large gradient Gaussians.

3.3.2. Frequency-adaptive perturbation magnitude

Going a step further, we adapt the perturbation magnitude according to the local image frequency. Specifically, for each Gaussian, we query the value of the optimal scale map at the projected center coordinate (x, y) on the 2D camera plane, $\gamma_i \leftarrow \Gamma^v(x, y)$. Since this value is obtained in the 2D image space, we multiply the rendered depth at the same location, $d_i \leftarrow D^v(x, y)$, and divide by the focal length

⁴We apply this adaptivity to the mean μ_i , rotation $\mathbf{q}_i$, and scale $\mathbf{s}_i$, which are geometric attributes of the Gaussian.

Table 1. **Quantitative comparison.** Our method improves baselines across the board.

Method	LLFF (3 views)			MipNeRF-360 (12 views)		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
3DGS [ACM ToG’23]	19.810 $\pm$.339	.6790 $\pm$.0078	.2145 $\pm$.0065	18.903 $\pm$.179	.5499 $\pm$.0036	.3734 $\pm$.0042
+ Ours	20.783 $\pm$.300	.7197 $\pm$.0032	.1965 $\pm$.0034	19.303 $\pm$.185	.5622 $\pm$.0051	.3552 $\pm$.0047
CoR-GS [ECCV’24]	20.185 $\pm$.142	.7015 $\pm$.0040	.2029 $\pm$.0035	19.515 $\pm$.243	.5733 $\pm$.0049	.3741 $\pm$.0066
+ Ours	20.862 $\pm$.154	.7283 $\pm$.0042	.1932 $\pm$.0030	19.805 $\pm$.233	.5833 $\pm$.0058	.3681 $\pm$.0069
DropGaussian [CVPR’25]	20.461 $\pm$.212	.7070 $\pm$.0045	.2064 $\pm$.0047	19.514 $\pm$.199	.5722 $\pm$.0042	.3657 $\pm$.0036
+ Ours	20.853 $\pm$.227	.7295 $\pm$.0045	.1969 $\pm$.0040	19.625 $\pm$.254	.5750 $\pm$.0053	.3627 $\pm$.0051
NexusGS [CVPR’25]	21.048 $\pm$.049	.7382 $\pm$.0008	.1776 $\pm$.0009	18.506 $\pm$.098	.5222 $\pm$.0031	.3587 $\pm$.0021
+ Ours	21.348 $\pm$.078	.7511 $\pm$.0012	.1714 $\pm$.0011	18.736 $\pm$.103	.5316 $\pm$.0034	.3522 $\pm$.0024
SE-GS [ICCV’25]	20.725 $\pm$.217	.7203 $\pm$.0049	.1861 $\pm$.0058	19.931 $\pm$.288	.5930 $\pm$.0063	.3702 $\pm$.0054
+ Ours	21.141 $\pm$.223	.7403 $\pm$.0042	.1803 $\pm$.0038	20.135 $\pm$.232	.5960 $\pm$.0059	.3644 $\pm$.0052

f to adjust it for the Gaussian mean μ_i and scale s_i , so that Gaussians located farther from the camera v are perturbed more strongly in 3D space. Then, the resulting value is multiplied by a predefined neighborhood radius to determine the final perturbation. Therefore, instead of Eq. (4) Algorithm 1.4 becomes

$$\hat{\theta}_i \leftarrow \theta_i + \gamma_i \frac{d_i}{f} \rho_{\theta} \frac{\nabla_{\theta_i} L}{\|\nabla_{\theta_i} L\|_2}. \quad (5)$$

As a result, the perturbation magnitude is adaptive to the frequency of the corresponding Gaussian. This adaptation mitigates the first-order approximation error of sharpness at Gaussians in high-frequency regions and perturbs strongly in low-frequency regions. Consequently, estimation of sharpness is more appropriate, thus improving generalization.

3.3.3. Frequency-adaptive sharpness weighting

Finally, we adapt the regularization weight according to the frequency of each Gaussian, where the frequency is defined the same as in Sec. 3.3.2. From Eq. (3), the final gradient is a weighted combination of the gradient at the original point and the gradient at the perturbed point:

$$\nabla_{\hat{\theta}_i} \hat{L} \leftarrow \frac{1 - 2\tilde{\gamma}_i}{1 - \tilde{\gamma}_i} \nabla_{\theta_i} L + \frac{\tilde{\gamma}_i}{1 - \tilde{\gamma}_i} \nabla_{\hat{\theta}_i} \hat{L}. \quad (6)$$

Here, $\tilde{\gamma}_i \leftarrow 0.95\gamma_i/\gamma_{\max}$, where $\gamma_{\max}$ is the maximum candidate scale of the LoG kernel, and 0.95 is determined empirically. We then update the parameter at the original point using this weighted gradient (Algorithm 1.8).

As a result, the sharpness penalty is reduced in high-frequency regions, preserving fine details, while applying a stronger penalty in low-frequency regions to facilitate better regularization.

4. Experiments

Dataset. We evaluate our method on LLFF [40] and MipNeRF-360 [5], following previous works [45, 63, 66, 68], where the input resolution is $8\times$ downsampled, and 3 and 12 input views are split for LLFF and MipNeRF-360, respectively.

Implementation. We choose the publicly available 3DGS [23] and its follow-up works as baselines. Specifically, we choose the state-of-the-art NexusGS [66], which leverages foundation models, and CoR-GS [63], DropGaussian [45], and SE-GS [65], which do not.

Metrics. We use PSNR, SSIM [55], and LPIPS [64] as evaluation metrics, where LPIPS is computed with a VGG network [50]. Additionally, we use Average Error (AVGE) [44], which is the geometric mean of PSNR, SSIM, and LPIPS. Considering the randomness of 3DGS, we conduct ten runs on the LLFF dataset and five runs on the MipNeRF-360 dataset.

4.1. Reconstruction quality

As shown in Tab. 1, our method achieves clear gains in all metrics, datasets, and baselines. Importantly, these improvements are achieved without any architectural changes or additional priors, but only with our optimization strategy. Fig. 4 show visual comparisons. Applying our method consistently improves the baselines by correcting geometric inaccuracies and reducing floating artifacts in novel view-points. These results reveal that our proposed optimization algorithm can be seamlessly integrated into current and future 3DGS-based frameworks, providing complementary enhancements.

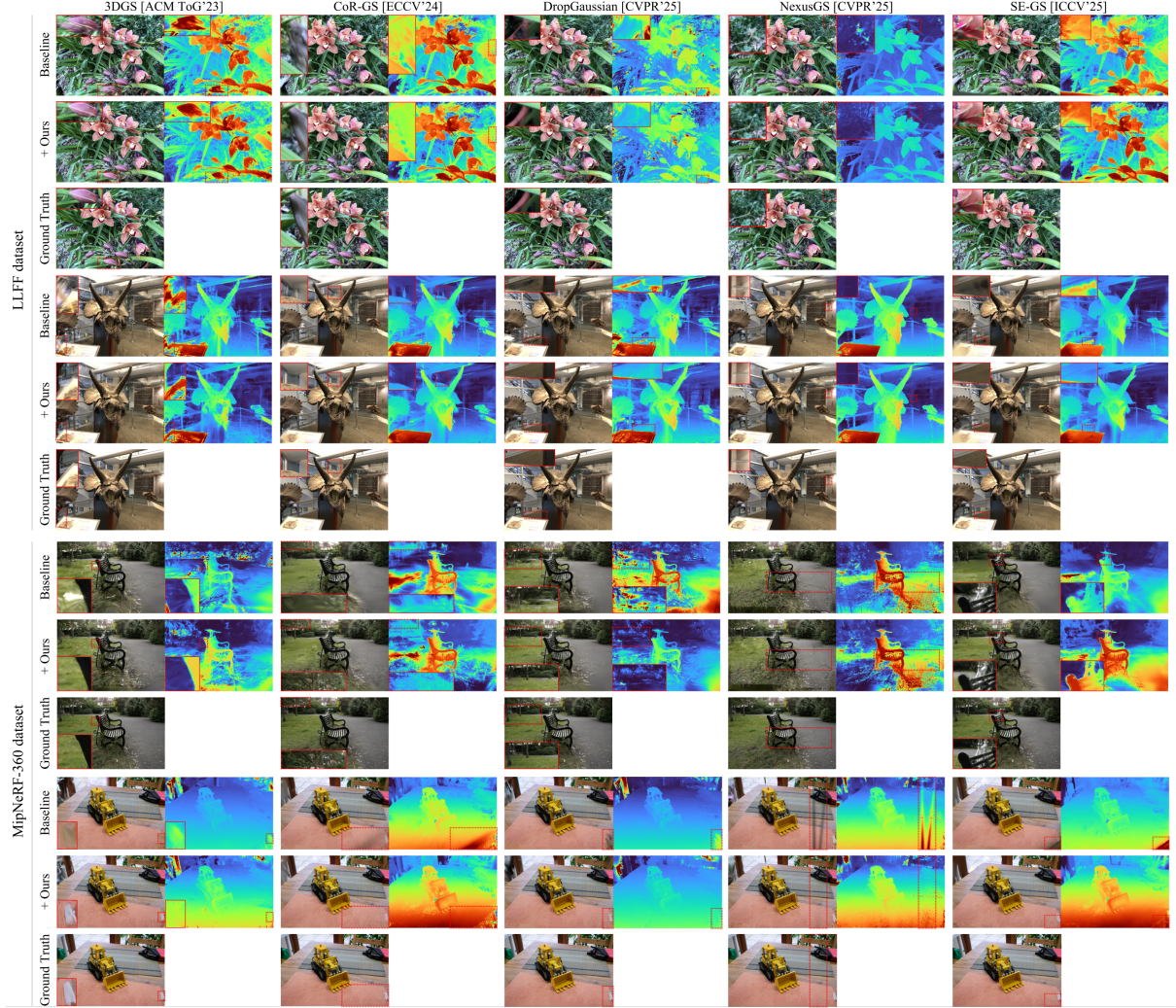


Figure 4. **Qualitative comparison.** Please zoom on the insets in red boxes to compare reconstruction quality.

4.2. Analysis

4.2.1. Ablation study

Directly applying SAM [12] to 3DGS leads to degraded performance. This naive approach strongly perturbs Gaussians with large gradients, resulting in blurry reconstructions (Fig. 5, second column). Moreover, it increases the first-order approximation error of sharpness estimation, hindering the optimization process of SAM (Tab. 2, fourth row). Separate sharpness per-Gaussian, handles each Gaussian separately. Thus, it mitigates this issue, leading to improvements in some metrics (Tab. 2, fifth row), but the results remain blurry (Fig. 5, third column). Frequency-adaptive sharpness weighting preserves sharpness in high-frequency details. Meanwhile, frequency-adaptive perturbation magnitude enables more faithful estimation by adaptively adjusting perturbation magnitude. Applying either compo-

nent individually shows performance gains (Tab. 2, sixth and seventh rows). However, removing either component degrades performance (Fig. 5, fourth and fifth columns). Using both achieves a better generalization, balancing between sharpness reduction and detail preservation (Fig. 5, sixth column; Tab. 2, eighth row).

4.2.2. Loss landscape visualization

To analyze the convergence behavior of our method, we visualize the reconstruction loss landscape and the corresponding optimization trajectories. We first train 3DGS for 5k iterations. Subsequently, we continue training for an additional 5k iterations with the densification process disabled, under three distinct settings: 3DGS optimization, SAM, and our proposed method. For visualization, we project the high-dimensional parameter trajectories onto a 2D plane using Principal Component Analysis with parameters from both trajectories.

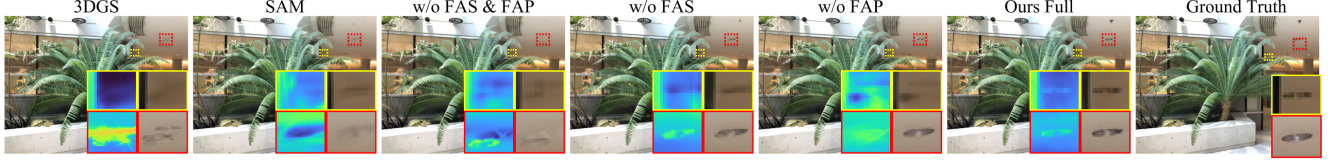


Figure 5. **Ablation study.** FAP and FAS denote frequency-adaptive perturbation magnitude and frequency-adaptive sharpness weighting, respectively. “3DGS”, “SAM”, and “w/o FAS & FAP” produce inaccurate geometry (red box). All except “Ours Full” show blurry results (yellow box).

Table 2. **Ablation study.** We report averaged results over ten runs. Standard deviations are omitted due to space constraints. SSG, FAP, and FAS denote separate sharpness per-Gaussian, frequency adaptive perturbation magnitude, and frequency adaptive sharpness weighting, respectively.

Components				LLFF (3 views)		
SAM	SSG	FAP	FAS	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
$\times$	$\times$	$\times$	$\times$	19.810	.6790	.2145
$\checkmark$	$\times$	$\times$	$\times$	20.198	.6958	.2095
$\checkmark$	$\checkmark$	$\times$	$\times$	20.390	.6977	.2174
$\checkmark$	$\checkmark$	$\times$	$\checkmark$	20.570	.6980	.2142
$\checkmark$	$\checkmark$	$\checkmark$	$\times$	20.560	.7116	.2023
$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	20.783	.7197	.1965

Table 3. **Performance by covisibility level on LLFF dataset.** Our method shows greater improvement with higher view sparsity.

Covisibility level	3DGS	AVGE $\downarrow$ + Ours	Δ
Covisibility 3	.0483 $\pm$.0093	.0417 $\pm$.0080	- .0066 $\pm$.0043
Covisibility 2	.0757 $\pm$.0156	.0644 $\pm$.0137	- .0114 $\pm$.0067
Covisibility 1	.0948 $\pm$.0256	.0806 $\pm$.0232	- .0142 $\pm$.0074

SAM converges to flatter minima than 3DGS. In Fig. 6a, the loss range between the local maximum and minimum is $2.37\times$, and the measured sharpness $\lambda_{\max}$ is $1.49\times$ smaller. SAM also achieves a test loss of 0.0129 lower than 3DGS, narrowing the generalization gap from 0.0985 to 0.0809. However, our method converges to less flat minima than SAM. As shown in Fig. 6b, the local loss range and measured sharpness of ours are $1.11\times$ smaller and $1.01\times$ smaller than 3DGS, respectively. Interestingly, our method achieves test loss of 0.0141 lower than 3DGS and further narrows the generalization gap to 0.0805, outperforming SAM in generalization.

These results suggest that sharpness is not strictly correlated with generalization, supporting our hypothesis that sharpness of high-frequency pixel should be preserved. This aligns with our finding (Fig. 5, second column) that SAM tends to over-penalize high-frequency details, leading

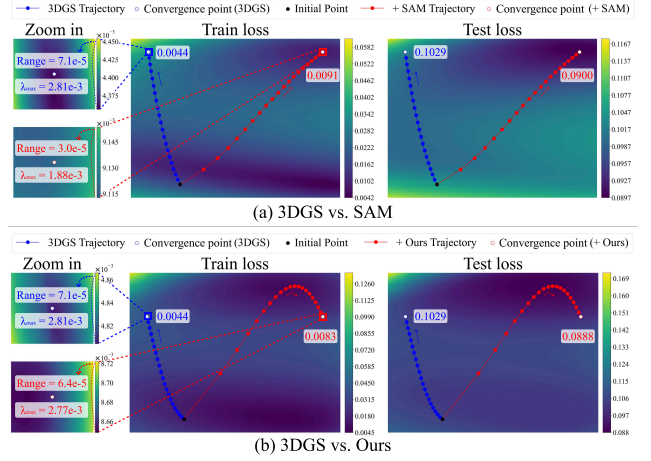


Figure 6. **Loss landscape visualization.** We compare the convergence behaviors of 3DGS, SAM, and Ours. Because the visualization produces a smoothed loss landscape, we provide a zoomed-in view near the convergence points. We measure sharpness as the maximum eigenvalue $\lambda_{\max}$ of the Hessian matrix [38, 57].

to blurry results.

4.2.3. Performance improvement by covisibility level

The improvement from our method is more substantial in regions observed by fewer training views. Following CoMapGS [20], we compute covisibility maps using MAST3R [30]. As shown in Tab. 3, the improvement of our method over the baseline 3DGS increases progressively as the covisibility decreases. This behavior aligns with our hypothesis that our method enhances generalization, particularly in under-constrained regions.

4.3. Application

4.3.1. Reducing computation cost of FASR

The main limitation of SAM and its follow-up work is that they compute the loss gradient twice at each step, theoretically doubling the training time. Fortunately, applying SAM on the last few training epochs improves performance similarly to the full application [67]. Based on this finding, we apply our method during the last 12.5% of the total iterations, denoted as Ours-L. As shown in Tab. 4, Ours-

Table 4. **Applying FASR at late training phase.** For efficient training, we apply our method during the later iterations, denoted as Ours-L. We report metrics (and their changes relative to 3DGS) averaged over ten runs on an RTX A5000; standard deviations are omitted for brevity. **Bold** indicates the best performance, and underline indicates the second best.

Method	LLFF (3 views)	
	AVGE ↓	Time (sec) ↓
3DGS	.1111	85.7
3DGS + Ours	.0979 (-.0132)	237. (2.77×)
3DGS + Ours-L	<u>.1016</u> (-.0095)	<u>98.2</u> (1.15×)

Table 5. **Applying FASR to online dynamic 3D Gaussians.** Our method is effective in dynamic scenarios under temporal sparsity, notably improving temporal consistency by reducing mTV.

Method	Neural 3D Video		
	PSNR ↑	SSIM ↑	mTV ↓
Yun et al. [SIGGRAPH'25]	32.542	.9486	.1109
+ Ours	32.622	.9497	.0989

L improves 3DGS by 0.0095 in AVGE, which are slightly (by 0.0037) less than full Ours. Moreover, compared to the baseline 3DGS, Ours increases the training time by 2.77×, Ours-L increases it by only 1.15×, making it more efficient.

The performance degradation of Ours-L occurs because 3DGS adaptively adjusts its number of learnable parameters through densification—a key difference from neural networks. This makes early training iterations influential, leading different results from Zhou et al. [67].

4.3.2. Improving generalization in temporal sparsity

Additionally, we extend our method to dynamic scenes captured with multi-view cameras, the Neural 3D Video dataset [33]. Specifically, we conduct experiments in an online configuration [14, 17, 32, 59], where observations are spatially dense but temporally sparse, meaning that we can only access the current frame in a sequentially processed video stream. We applied our method to Yun et al. [62] with 3DGStream [52] backbone. Following their protocol, we select the first frame 3D Gaussians with the highest PSNR for initialization and compute the masked total variation (mTV) to measure temporal consistency.

Our method improves temporal consistency and visual quality compared to the baseline (Tab. 5). This shows that our approach enhances generalization not only in the spatial domain but also in the temporal domain. Furthermore, Yun et al. [62] claims that one cause of temporal jittering is the inevitable noise in training datasets. Since SAM shows robustness on noisy training data, our finding aligns with this explanation.



Figure 7. **Qualitative comparison with FreeNeRF.** Please zoom on the insets in red boxes to compare reconstruction quality.

Table 6. **Quantitative comparison with FreeNeRF.** Our approach improves the NeRF baseline.

Method	LLFF (3 views)		
	PSNR ↑	SSIM ↑	LPIPS ↓
FreeNeRF [CVPR'23]	19.523	.6063	.3103
+ Ours	19.584	.6182	.2983

4.3.3. Applying our intuition to NeRF

Despite our method is not directly designed for implicit representations such as NeRF [41] because a single parameter affects all pixels, we can alternatively apply our intuition to baselines that learn in a coarse-to-fine procedure [36, 60]. Specifically, we set a strong perturbation magnitude and regularization weight when the NeRF parameters learn low-frequency components, and gradually decrease these values as the model learns high-frequency details.

We applied this approach to FreeNeRF [60] and demonstrate that it improves both visual quality in novel view synthesis (Fig. 7) and quantitative results (Tab. 6). Although the improvement over the NeRF baseline is smaller than that in the 3DGS baselines due to discrepancy in the representation, these additional gains still support our intuition and versatility across different representations.

5. Conclusion

In this work, we present the first fundamental investigation linking loss landscape and generalization to novel view synthesis, thereby improving quality of 3D Gaussian Splatting [23], especially in sparse view reconstruction. We propose Frequency-Adaptive Sharpness Regularization (FASR), an optimization algorithm that reformulates Sharpness-Aware Minimization (SAM) [12] in a frequency-adaptive manner. This reformulation overcomes the limitation of SAM in reconstruction tasks, achieving generalization as well as fine detail reconstruction. FASR is easily applicable across diverse baselines and can be further extended to NeRF-based models and dynamic scenes in temporally sparse scenarios. We hope this work inspires the research community to explore the link between sharpness and generalization in reconstruction fields.

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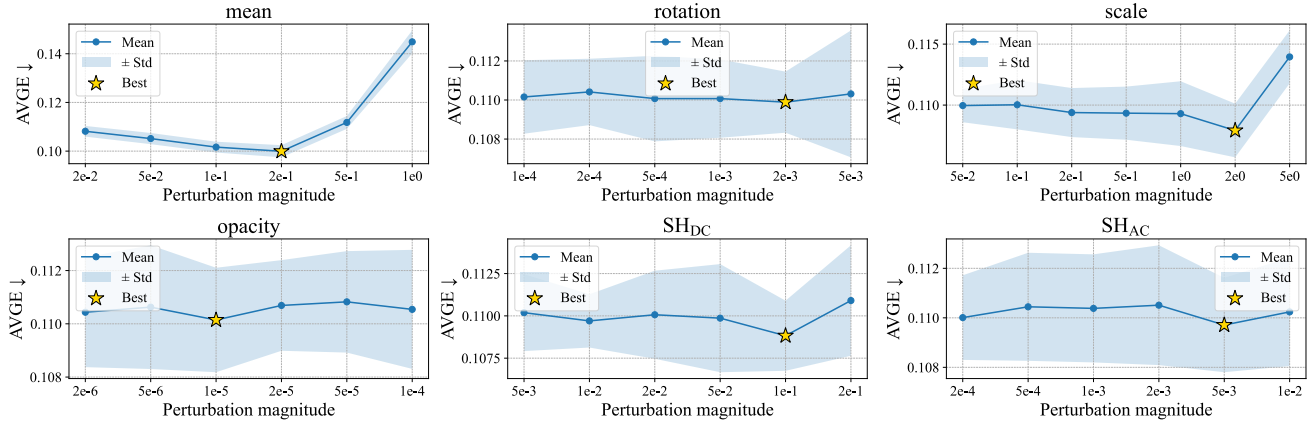


Figure 8. **Hyperparameter grid search, where each parameter is perturbed individually on the LLFF dataset.** Plots are means and standard deviations over ten runs. We mark a star at the best hyperparameter value.

A. Hyperparameter search

Selecting the optimal neighborhood of radius ρ , i.e., the perturbation magnitude, remains a challenging problem in SAM-based methods. Although our method scales ρ for each Gaussian considering local image frequency, ρ itself remains a core hyperparameter. Similar to SAM, we first grid search the optimal perturbation magnitude by perturbing each parameter individually, as shown in Fig. 8. However, the optimal ρ when perturbing all parameters simultaneously is typically smaller than the values found in this search because the parameters are not independent and mutually influence each other. Therefore, we select candidates including the individually found ρ and smaller values, and then perform a grid search with simultaneous perturbation to determine the final hyperparameter.

B. Contribution of each Gaussian attribute

To analyze the contribution of each Gaussian attribute to the overall performance, we apply our method either exclusively to a single attribute or to all attributes except one. Nevertheless, all attributes contribute positively to the performance, the Gaussian mean is the dominant contributor to the performance gain (Tab. 7). This finding suggests that instead of perturbing all attributes simultaneously, applying our method exclusively to the Gaussian mean could be a simplified approach, significantly reducing hyperparameter search complexity while likely retaining a substantial portion of the performance improvement.

C. Comparison to random perturbation

To support our claims in Sec. 2.3, we demonstrate the impact of applying random perturbations to Gaussian parameters during training. As shown in Fig. 9, random perturba-

Table 7. **Ablation study on the contribution of each Gaussian attributes.** The Gaussian mean is the dominant contributor to the performance improvement.

Method	LLFF (3 views)		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
3DGS + Ours	20.783 $\pm$.300	.7197 $\pm$.0032	.1965 $\pm$.0034
w/o mean	20.053 $\pm$.228	.6903 $\pm$.0051	.2102 $\pm$.0040
w/o rotation	20.656 $\pm$.241	.7175 $\pm$.0046	.1976 $\pm$.0040
w/o scale	20.641 $\pm$.235	.7171 $\pm$.0049	.1987 $\pm$.0035
w/o opacity	20.628 $\pm$.218	.7175 $\pm$.0044	.1977 $\pm$.0033
w/o SH _{DC}	20.594 $\pm$.242	.7152 $\pm$.0052	.1972 $\pm$.0036
w/o SH _{AC}	20.599 $\pm$.205	.7169 $\pm$.0041	.1973 $\pm$.0034
w/ mean	20.561 $\pm$.268	.7147 $\pm$.0046	.1975 $\pm$.0037
w/ rotation	19.854 $\pm$.168	.6828 $\pm$.0029	.2110 $\pm$.0024
w/ scale	20.036 $\pm$.222	.6871 $\pm$.0048	.2093 $\pm$.0037
w/ opacity	19.888 $\pm$.188	.6803 $\pm$.0045	.2124 $\pm$.0033
w/ SH _{DC}	19.990 $\pm$.202	.6867 $\pm$.0048	.2115 $\pm$.0034
w/ SH _{AC}	19.914 $\pm$.186	.6827 $\pm$.0042	.2114 $\pm$.0033
3DGS	19.810 $\pm$.339	.6790 $\pm$.0078	.2145 $\pm$.0065

tions often introduce unexpected artifacts and yield smaller performance gains than ours (Tab. 8).

Furthermore, to demonstrate the importance of finding a local maximum by adversarial perturbation, we compare our method with a variant that randomly samples parameters within the neighborhood radius. As shown in Tab. 8, the improvement of random perturbation is smaller than ours, indicating that adversarially perturbing in the gradient direction is essential for effective sharpness regularization.

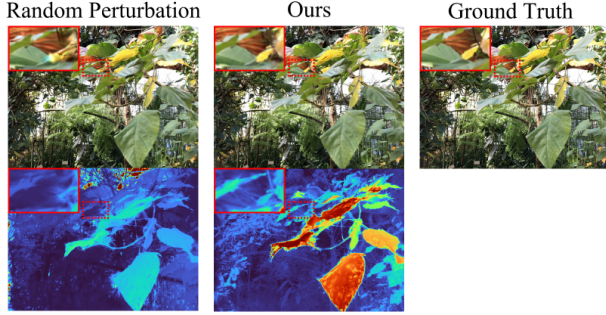


Figure 9. **Qualitative comparison of random perturbation and our method.** Random perturbation often introduce unexpected artifacts.

Method	LLFF (3 views)		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
3DGS	19.810 $\pm$.339	.6790 $\pm$.0078	.2145 $\pm$.0065
w/ RP	20.174 $\pm$.207	.6987 $\pm$.0040	.2009 $\pm$.0032
w/o AP	20.263 $\pm$.289	.6990 $\pm$.0031	.2056 $\pm$.0027
3DGS + Ours	20.783 $\pm$.300	.7197 $\pm$.0032	.1965 $\pm$.0034

Table 8. **Quantitative comparison of random perturbation and adversarial perturbation.** RP and AP denote random perturbation on Gaussian parameters and adversarial perturbation, respectively

D. Analysis of late-phase application

We provide a detailed study on the application timing of our method. In Sec. 4.3.1, we follow Zhou et al. [67], applying our method only during the final 12.5% of the total training iterations. Extending this, we further analyze the impact of the starting point by varying the application duration in 10% increments of the total iterations. As shown in Fig. 10, fully applying our method yields the best performance, and the performance degrades as the application phase is later in the training process.

E. LoG for local image frequency estimation.

We calculate the local frequency at each pixel of the input image employing a multi-scale analysis inspired by blob detection using the Laplacian of Gaussian (LoG) [35]. For the grayscale image, we calculate the absolute LoG response at each candidate scale. We capture the optimal scale at each pixel by finding the first significant increase in its response curve that exceeds the threshold. The optimal scale is the smallest one that precedes such a rise; otherwise, we assign the maximum candidate scale. This process results in an optimal scale map $\Gamma^v \in \mathbb{R}^{H \times W}$ at view v , with H and W denoting the height and width of the image, respectively. The local frequency is inversely proportional to this selected

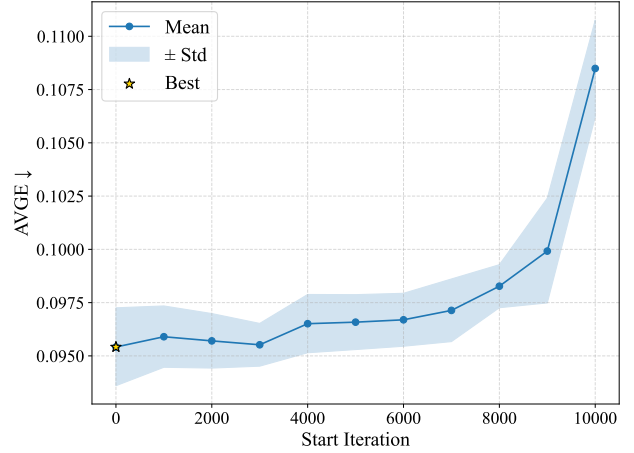


Figure 10. **Ablation study on the application phase.** We demonstrate the performance improvements obtained by varying the application duration in 10% increments of the total training iterations.

scale.

F. Implementation detail

For all methods except 3DGS, we use the official repository. Since the official 3DGS does not target sparse-view reconstruction, we refactor it by referring to CoRGS and DropGaussian. NexusGS does not provide optical flow for MipNeRF-360; we compute the flow using FlowFormer++ [49]. When the baseline utilizes regularization, we backpropagate it after computing the gradient of FASR. When applying our method in the deformation process, we compute the FASR gradient on the deformed Gaussians and propagate it back to the deformation network. We update the residual map of Yun et al. [62] during the ascent step and the Gaussians during the descent step. For each dataset, we use the same values of ρ_θ and γ for all baselines except NexusGS, where we set them to $0.1 \times$ the values used for other baselines due to its use of denser per-pixel Gaussians.

G. More results

We report the quantitative results of each scene in Tab. 9. Our method method outperforms the baseline in most cases.

Table 9. **Per-scene quantitative results on LLFF and MipNeRF-360 dataset.** We report AVGE (lower is better) of each scene.

Method	LLFF (3 views)							
	fern	flower	fortress	horns	leaves	orchids	room	trex
3DGS	.0950 $\pm$.01790	.1147 $\pm$.00180	.0885 $\pm$.00390	.1276 $\pm$.00240	.1362 $\pm$.00170	.1769 $\pm$.00190	.0753 $\pm$.00160	.0749 $\pm$.00210
+ Ours	.0765 $\pm$.00080	.1066 $\pm$.00140	.0781 $\pm$.00530	.1044 $\pm$.00280	.1170 $\pm$.00130	.1605 $\pm$.00190	.0686 $\pm$.00190	.0712 $\pm$.00710
CoR-GS	.0806 $\pm$.00080	.1088 $\pm$.00290	.0754 $\pm$.00160	.1205 $\pm$.00120	.1427 $\pm$.00310	.1735 $\pm$.00200	.0723 $\pm$.00130	.0699 $\pm$.00440
+ Ours	.0724 $\pm$.00090	.1014 $\pm$.00220	.0683 $\pm$.00190	.1075 $\pm$.00200	.1326 $\pm$.00240	.1603 $\pm$.00100	.0695 $\pm$.00100	.0614 $\pm$.00340
DropGaussian	.0791 $\pm$.00120	.1056 $\pm$.00210	.0796 $\pm$.00420	.1186 $\pm$.00270	.1284 $\pm$.00190	.1650 $\pm$.00160	.0715 $\pm$.00150	.0687 $\pm$.00190
+ Ours	.0717 $\pm$.00100	.1048 $\pm$.00170	.0784 $\pm$.00340	.1101 $\pm$.00340	.1176 $\pm$.00140	.1536 $\pm$.00170	.0701 $\pm$.00200	.0657 $\pm$.00260
NexusGS	.0893 $\pm$.00030	.0981 $\pm$.00020	.0498 $\pm$.00030	.0909 $\pm$.00050	.1072 $\pm$.00030	.1383 $\pm$.00060	.0755 $\pm$.00060	.0810 $\pm$.00050
+ Ours	.0848 $\pm$.00050	.0989 $\pm$.00060	.0475 $\pm$.00090	.0897 $\pm$.00070	.0993 $\pm$.00060	.1354 $\pm$.00040	.0696 $\pm$.00030	.0741 $\pm$.00060
SE-GS	.0720 $\pm$.00040	.1052 $\pm$.00180	.0662 $\pm$.00130	.1104 $\pm$.00180	.1392 $\pm$.00230	.1707 $\pm$.01440	.0635 $\pm$.00220	.0596 $\pm$.00180
+ Ours	.0675 $\pm$.00060	.0992 $\pm$.00160	.0661 $\pm$.00820	.1028 $\pm$.00160	.1310 $\pm$.00160	.1546 $\pm$.00170	.0629 $\pm$.00120	.0564 $\pm$.00080
Method	MipNeRF-360 (12 views)							
	bicycle	bonsai	counter	garden	kitchen	room	stump	
3DGS	.1763 $\pm$.00400	.1408 $\pm$.00400	.1464 $\pm$.00250	.1302 $\pm$.00190	.1052 $\pm$.00160	.1078 $\pm$.00110	.2347 $\pm$.00280	
+ Ours	.1714 $\pm$.00120	.1285 $\pm$.00240	.1350 $\pm$.00110	.1240 $\pm$.00130	.1017 $\pm$.00160	.1021 $\pm$.00460	.2253 $\pm$.00640	
CoR-GS	.1734 $\pm$.00340	.1262 $\pm$.00250	.1310 $\pm$.00120	.1297 $\pm$.00370	.1054 $\pm$.00360	.0931 $\pm$.00230	.2214 $\pm$.00770	
+ Ours	.1733 $\pm$.00170	.1131 $\pm$.00230	.1229 $\pm$.00060	.1287 $\pm$.00290	.1044 $\pm$.00320	.0993 $\pm$.00440	.2073 $\pm$.00950	
DropGaussian	.1646 $\pm$.00110	.1301 $\pm$.00210	.1356 $\pm$.00140	.1243 $\pm$.00110	.0979 $\pm$.00270	.1026 $\pm$.00500	.2187 $\pm$.00410	
+ Ours	.1613 $\pm$.00310	.1286 $\pm$.00070	.1309 $\pm$.00100	.1218 $\pm$.00200	.1027 $\pm$.00210	.1003 $\pm$.00440	.2171 $\pm$.01180	
NexusGS	.1818 $\pm$.00100	.1308 $\pm$.00190	.1563 $\pm$.00130	.1249 $\pm$.00030	.0941 $\pm$.00070	.1506 $\pm$.00380	.2352 $\pm$.00240	
+ Ours	.1739 $\pm$.00100	.1278 $\pm$.00060	.1531 $\pm$.00160	.1243 $\pm$.00040	.0921 $\pm$.00090	.1421 $\pm$.00450	.2299 $\pm$.00210	
SE-GS	.1624 $\pm$.00350	.1179 $\pm$.00300	.1251 $\pm$.00370	.1207 $\pm$.00160	.1107 $\pm$.00370	.0875 $\pm$.00380	.2014 $\pm$.00830	
+ Ours	.1599 $\pm$.00690	.1105 $\pm$.00300	.1217 $\pm$.00170	.1179 $\pm$.00250	.1092 $\pm$.00350	.0973 $\pm$.00090	.1941 $\pm$.00480	